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Which Arduino should you go for?

The selection of boards can be done based on design, investment and support requirements. The performance of a board depends on the customisation that is being done by the programmer and architect of the project. eCommerce giants like eBay and Amazon offer a lot of compatible models on their sites.

Experts recommend Arduino UNO, Arduino Mega or Arduino Mini for starters. All their specifications are listed on the Arduino's website. The developer can also access an array of information that supports the project.

Arduino UNO. There are two variants to this version of Arduino: the normal version and the surface mount device (SMD) version. The normal version has the integrated chip in a socket, while the SMD version has an ATmega328 controller soldered onto the board. The advantage here is that, any damage to the IC in the normal version can be rectified with just replacement of the chip, whereas the whole board has to be replaced in case of SMD Arduino.

Arduino Mini. The boards of Arduino Mini are smaller in size, with a length of about three centimetres only, and are cheaper. The Arduino Mini is packed with functionalities similar to that of an Arduino UNO. It is best suited for small robots and even smaller products. The only problem here is that the integrated chip is mounted on the board and, like an SMD board, the whole board has to be replaced in case of problem with any of its components.

Arduino Mega. The Mega board from Arduino can accommodate projects that require more than 20 GPIO pins. It can accommodate a lot of wiring. A simple addition of IC 74595 can also help increase the number of pins on an Arduino Uno, but buying an Arduino Mega would be the best option. The Arduino Mega adds two more pins to existing two pairs of Tx and Rx pins than its other variants.

Arduino Lilypad. This Arduino variant is used for more creative projects due to its sleek design. The peculiarity of Lilypad is that here wires are replaced by conductive threads, which can be easily stitched into clothing and fabric. Projects for special purposes and smart clothing can be designed with a Lilypad board. You can look forward to some creative projects with this board.

In nutshell. To summarise, an UNO or Mini can be chosen for small and simple Arduino projects depending on the size and cost limitations. Or, if you need a greater number of pins or serial communication devices to be processed with the same controller, you can go for Arduino Mega. For a beginner, Arduino UNO is the most recommended board to get started with. Packing about 8MHz CPU clock speed, it houses ATmega328 or an ATmega32u4 processor.

port device and ports with respect to the Arduino board you have in hand. Once the port is connected and selected from the software, know that the board is ready to uplink the sketch; you will see Rx or Tx LEDs blinking on your board while the sketch is being uploaded. The Arduino bootloader is a small program that facilitates your program to be loaded onto the board. The bootloader also blinks while it is uploading the most recently uploaded sketch.

As a programmer it is important to know the built-in functions. Libraries come in handy once a programmer decides to use

extended abilities. Typical programs require one of the *#include* statements so that the programs can run with more agility. Though this increases the space consumption, it reduces complexity of the program and makes it easy to connect to a sensor, display modules, etc. Additional libraries can be downloaded from Library Manager while some are included in Arduino itself.

The latest version of the Arduino software, version 1.6.8, is included in this month's DVD. **EFY**

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